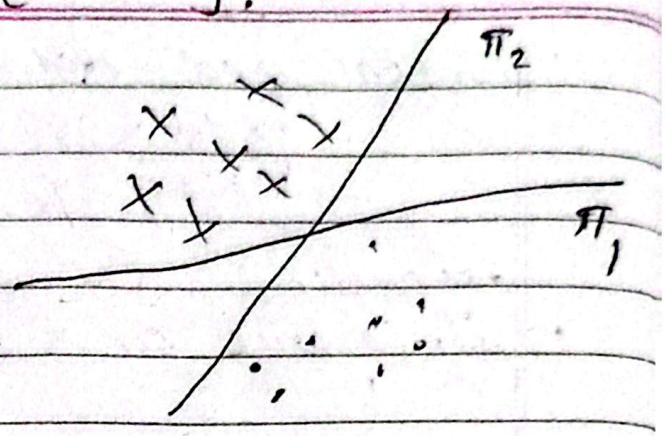
Support Vector Machines [SVM].

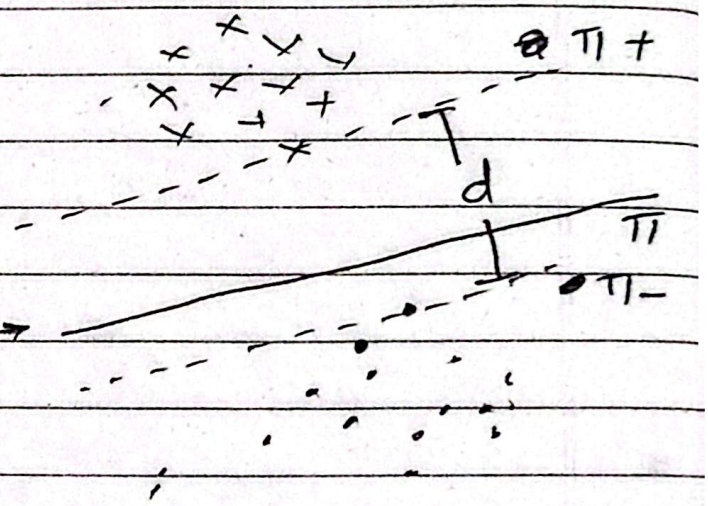
$\sigma(w^T x_i) = P(y=-1)$



- SVM wants to take atmost distance [margin] betn point & hyperplane. better

margin → maximizing hyperplane.

- random select hyperplane.
- parallel to hyper plane but should be nearest point in both sides [$\pi+$ & $\pi-$]

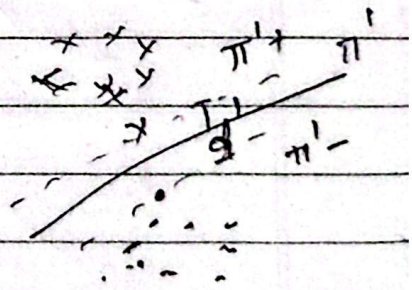


randomly selected.

Now, calculate the margin.

margin [d] → need max value.

- Again, do same for all.



$d > d'$

- Hence choose 'd'.

$\hat{y} = w^T x + b$

for which d is maximum.

Disadvantage:
- not larger datasets
- overlapping classes

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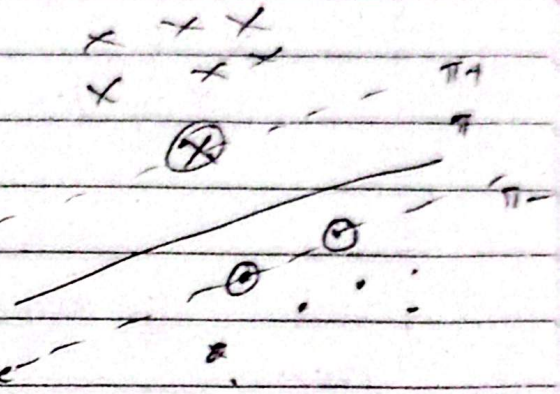
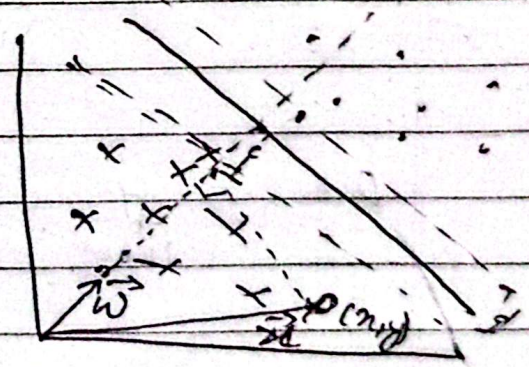
- Stop after getting 1st positive & negative points, [called support vectors] ⊗ & two ⊙ are support vectors.



- hyperplane should be at most distance from support vectors.

Advantages:
- SVM are stronger robust to outliers.
- Non-Linear data can also be solved by SVM.
- Both classification & Regression.
- Work with high dimensional data.

Mathematics of SVM. [Hard margin SVM] (we need perfect data without one mistake)



- Here, $\vec{w}$ is the vector
  - direction: perpendicular to 3 [π, π+, π−]
  - magnitude: unknown
- (x, y) $\vec{u}$ unknown vector whose classification is to be done.
- Take the projection of $\vec{u}$ in $\vec{w}$.

$$\vec{w}\cdot\vec{x} \geq c$$

or; $\vec{w}\cdot\vec{x} - c \geq 0$

or; $\vec{w}\,\vec{x} + b \geq 0$

If true then +ve

If false the −ve point.

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Let 2 D:

eq^n of line: $\underset{w_1}{2}x + \underset{w_2}{3}y + \underset{b}{3} = 0$.

Let we have to find of point (3,2),

then;

$\vec{w} = (2,3)$ $\quad$ $\vec{x} = (3,2)$

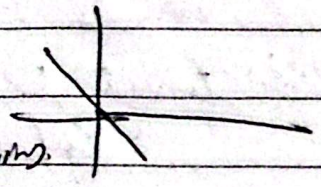
$w^T x$.

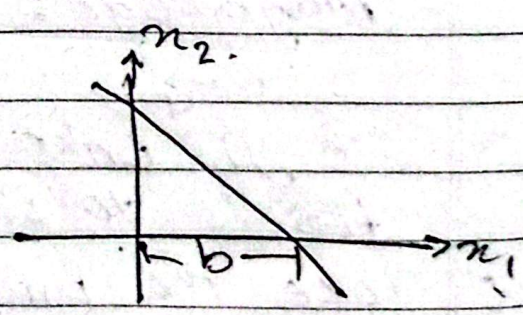
For any $\vec{x}_i$

$$\hat{y} = \begin{cases} +1 & \text{if } \vec{w}.\vec{x}_i + b \geq 0. \\ -1 & \text{if } \vec{w}.\vec{x}_i - b < 0. \end{cases}$$

Q. $\underset{\text{label}}{y} = w^T x$

[if pass through origin]



$y = w^T x + b$

distance from origin to x-intercept.